

Benudhar Behera

Lead Software Engineer | Java, Distributed Systems, Cloud & AI Platforms

Bangalore, India | +91- 9938636246 | benudharbehera003@gmail.com

LinkedIn: [linkedin.com/in/benudhar](https://www.linkedin.com/in/benudhar) | GitHub: github.com/benudhar

PROFESSIONAL SUMMARY

Lead Backend Engineer with **8.7+ years** of experience architecting high-performance distributed systems and cloud-native applications. Specialized in **Java, Spring Boot, Kafka, AWS, and event-driven microservices**, with hands-on expertise in **GenAI, LLMs, RAG, and Vector Databases**. Passionate about building scalable, resilient, and secure backend platforms through modern architecture, performance engineering, and cloud-native design principles.

TECHNICAL SKILLS

Programming Languages: Java (Primary), Python

Framework & Tools: Spring Boot, Spring WebFlux, FastAPI, REST APIs, Apache Kafka, Microservices

Database: PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (EC2, S3, RDS, EKS, Lambda, SQS, SNS), Terraform, Docker, Kubernetes, Helm, CI/CD, Jenkins

AI & GenAI: RAG, LangChain, OpenAI API, Vector Databases, Prompt Engineering, AI Orchestration, MLflow

Observability: Prometheus, Grafana, ELK Stack, Splunk, AWS CloudWatch, Distributed Tracing

Design Principles: System Design, HLD, LLD, Scalability, Clean Architecture, Design Patterns, SOLID, Rate Limiting

Process & Delivery: Agile, Scrum, Sprint Planning, Stakeholder Management, Cross-functional Collaboration

KEY ACHIEVEMENTS

- **Architected and delivered** a production-grade **GenAI platform** using **LLMs, RAG, and vector search**, supporting **70+ user requests/minute** with low-latency, secure, and highly available document retrieval.
- **Designed and led cloud-native event-driven microservices** using **Java 21, Spring Boot, Kafka, and AWS**, delivering scalable, resilient, and fault-tolerant backend services for enterprise applications.
- **Reduced API latency by ~40% through caching, query optimization, JVM tuning, and asynchronous processing**, significantly improving system throughput and application responsiveness.
- Built a **centralized authentication and authorization framework** using **OAuth2, OIDC, Keycloak, JWT, and RBAC**, enabling secure, centralized access control across distributed microservices.

PROFESSIONAL EXPERIENCE

Lead Backend Engineer (Senior Technology Consultant) | Ernst & Young (EY)

Bangalore, India • Mar 2024 - Present

- Led end-to-end architecture and delivery of an **enterprise-grade GenAI conversational platform**, integrating LLMs with large-scale research datasets to enable real-time intelligent insights.
- Designed and developed cloud-native microservices using **Java 21, Spring Boot and Kafka** following Hexagonal Architecture, enabling scalable and maintainable enterprise applications.
- Implemented **RAG-based knowledge retrieval system** using vector embeddings, significantly improving contextual response accuracy.
- **Built a RAG-based knowledge retrieval engine** using vector embeddings and semantic search, significantly improving contextual response accuracy and retrieval efficiency.
- **Developed** AI-powered conversational workflows for company research and enterprise data analysis, enabling faster and more accurate decision-making.
- **Collaborated** with product managers, AI engineers, and data teams to deliver secure, production-ready AI solutions using Agile delivery practices.

Senior Backend Engineer (Senior Project Engineer) | Wipro Limited

Bangalore, India • Jul 2021 – Feb 2024

- Designed and developed **cloud-native microservices** using **Java, Spring Boot and Kafka** for a large-scale enterprise platform, enabling scalable backend services and real-time event processing.
- Improved API response time by over **30% through query optimization**, caching strategies, asynchronous processing and JVM tuning.
- Optimized database **indexing strategies and SQL execution** plans to improve throughput for high-volume transactional workloads.
- Built comprehensive unit and integration test suites using **JUnit 5 and Mockito**, improving application stability and reducing production defects.
- Troubleshoot production incidents using **Splunk, Grafana and AWS CloudWatch**, reducing mean time to resolution through proactive monitoring and log analysis.

Backend Engineer (Associate IT Consultant) | ITC InfoTech

Bangalore, India • Sep 2020 – Jul 2021

- Developed **scalable cloud-native backend services** using **Java, Spring Boot and RESTful APIs** for an enterprise e-commerce platform, supporting order lifecycle management, payment processing and real-time transaction workflows
- Implemented **event-driven architecture using Kafka** to support asynchronous order processing, payment notifications and inventory synchronization across distributed services.
- Designed **idempotent APIs and resilient payment workflows** using retry and failure recovery mechanisms, ensuring reliable transaction processing.
- Improved API performance through **database optimization, indexing strategies and efficient query design**, reducing response latency for business-critical operations.
- Developed reusable business components following **SOLID principles**, clean architecture and design patterns, improving code quality and maintainability.
- Implemented comprehensive automated testing using JUnit 5 and Mockito, increasing **test coverage and ensuring production stability**.
- Participated in technical **design discussions, code reviews and production deployments**, contributing to the successful delivery of scalable enterprise applications.

Software Engineer | Silicon Tech Lab Pvt Ltd.

Bhubaneswar, India • Nov 2017 – Sep 2020

- Developed backend modules using **Java, Spring MVC, Hibernate ORM and PostgreSQL** for enterprise ERP applications supporting business process automation.
- Implemented **business logic and reusable service components** to support core ERP functionalities, improving application maintainability and reducing development effort.
- Resolved production issues through root cause analysis, bug fixing and performance tuning, improving application stability and customer satisfaction.
- Collaborated with senior engineers**, QA teams and business analysts to deliver production-ready features following Agile development practices.

CERTIFICATIONS

- EY IT Architecture - Solution Architecture (Bronze), 2025
- EY Artificial Intelligence - Applied AI (Bronze), 2024
- Introduction to Generative AI - Google Cloud

EDUCATIONAL QUALIFICATION

Bachelor of Technology (B. Tech) — Computer Science & Engineering

Silicon Institute of Technology, Bhubaneswar | 2013 – 2017